

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eight Semester of B. Tech. Examination (EE)

May 2014

EE417 Artificial Intelligence Techniques to Energy System

Date: 12.05.2014, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain different operations which could be performed on fuzzy sets. [07]
- Q - 2 (a) Explain different types of linguistic hedges with mathematical expression and graphical representation. [07]
- (b) Explain different types of membership functions used in fuzzy logic. [07]

OR

- Q - 2 (a) Three variables of interest in power transistor are the amount of current that can be switched, the voltage that can be switched and the cost. [07]
Average current(in Ampere)= $I = \{0.4/0.8, 0.7/0.9, 1/1, 0.8/1.1, 0.6/1.2\}$
Average Voltage(in Volt)= $V = \{0.2/30, 0.8/45, 1/60, 0.9/75, 0.7/90\}$
These two fuzzy sets are related to the power of transistor. Power in electronics is defined by an algebraic operation $P=VI$, but let us deal with a general Cartesian relationship between voltage and current, i.e. $P=V \times I$.
(a) Find $P=V \times I$. Now fuzzy set for the cost C in dollars, $C = \{0.4/0.5, 1/0.6, 0.5/0.7\}$
(b) Find $T=I \times C$.
(c) Using max-min composition, find $E=PoT$.
(d) Using max-product composition, find $E=PoT$.
- (b) Write down steps for the computational algorithm of reactive power control and draw the flow chart for reactive power control using fuzzy logic controller. [07]
- Q - 3 (a) Explain gradual steps to construct the neural networks for performing certain tasks. [03]
- (b) Give the classification of neural network learning algorithms. [04]
- (c) Explain the types of learning methods for ANN. [07]

OR

- Q - 3 (a) Explain various rules utilized for supervised and unsupervised learning method. [07]
- (b) With neat diagram explain Rosenblatt's perceptron. [05]
- (c) Can perceptron be trained by XOR function? Justify. [02]

SECTION – II

Q - 4 (a) Define: Optimization. Differentiate classical methods from Artificial Intelligence methods. [06]

(b) Using graphical method, find the values of x_1 and x_2 so as to maximize the following objective function [08]

$$Z=2x_1-3x_2$$

$$\text{s.t. } x_1+x_2 \leq 15, x_1-x_2 \geq 0, x_1 \leq 20, x_2 \geq 3, x_2 \leq 12, x_1, x_2 \geq 0$$

Q - 5 (a) Minimize following objective function using Steepest descent method. [06]

$$F(X)=4x_1^2+6x_2^2-6x_1x_2-4x_1. \text{ Perform one iteration.}$$

(b) Minimize the following objective function using Particle Swarm Optimization method [08]
OR Genetic algorithm. Perform only one iteration. Take particle size=04. Assume suitable values of remaining variables.

$$\text{Min. } z=2x_1+x_2$$

$$\text{s.t. } -4x_1-3x_2+6 \leq 0$$

Q - 6 (a) Explain optimal reactive power dispatch and its associated constraints. [05]

(b) Explain ill conditioning of the matrix. [02]
